TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Fibonacci ML SoC** **Hierarchically Scalable Machine Learning System-on-Chip for Ultra-Low-Power Edge Processing**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (.), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (.), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Semiconductor Systems		Bio and Medical Devices	
	Medium Category	Small Category	Medium Category	Small Category
	AI Processors	Edge AI System-on-Chip (SoC)	Embedded Artificial Intelligence	Low-Power Machine Learning Hardware
Concept of Proposed Technology	The **Fibonacci ML SoC** is a **machine-learning system-on-chip (SoC)** designed around the principle of **hierarchical scalability**, inspired by the Fibonacci number series. Similar to the Fibonacci sequence, where each element is formed by combining previous ones, the SoC can **dynamically increase its computational performance** by activating additional accelerator resources according to application needs. The heterogeneous architecture integrates: • a **low-power time-series ML accelerator (FETA)**, • **two clusters of highly parallelized neural processing units (NPUs)**, • **energy-optimized on-chip memories**, • a **flexible RISC-V microcontroller core**, and • a **rich set of peripherals** for system integration. Trained ML models can be deployed through **CSEM's ML compiler**, supporting common formats such as **ONNX**.			
Rationale for Joint R&D Project	Edge AI applications increasingly require: • simultaneous processing of spatial and temporal data, • dynamic adaptation of computational complexity, • and strict power efficiency across wide operating ranges. Korean SMEs developing edge AI products can benefit from a **hierarchically scalable ML SoC** that supports: • multimodal data processing, • staged evaluation with increasing complexity, • and dynamic power management.			

	Joint R&D is required to: • adapt the Fibonacci ML SoC architecture to application-specific requirements, • integrate it into SME product environments, • and validate performance and power efficiency under real workloads.
Key Words	1. Edge AI SoC 2. Hierarchical scalability 3. Neural processing unit (NPU) 4. Time-series ML accelerator 5. Ultra-low-power machine learning

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 5-6
R&D Objectives	• To implement a hierarchically scalable ML SoC architecture for edge processing. • To support spatial and temporal neural network workloads on a single SoC. • To enable dynamic activation of accelerator resources to optimize power consumption. • To demonstrate multimodal and multi-stage ML processing on embedded hardware.			

R&D Target Specifications	• **NPU Clusters**
	  ○ Optimized for spatial neural networks (CNNs, ResNets, MobileNets)
	  ○ Sparsity exploitation
	• **FETA Cluster**
	  ○ Optimized for temporal neural networks (LSTM, GRU)
	  ○ Integrated temporal feature extraction engine
	• **System Characteristics**
	  ○ Power consumption range: **100 µW – 500 mW**
	  ○ Support for multi neural network execution
	  ○ Selective execution and early exit mechanisms			

R&D Plan	**Phase 1 – Architecture Refinement and Toolchain Integration** • Refinement of hierarchical scalability mechanisms • Integration with ML compiler and ONNX model deployment • Mapping of workloads to NPU and FETA clusters **Phase 2 – SoC Integration and Embedded Validation** • Integration of RISC-V core, memories, and peripherals • Validation of dynamic precision scaling and power switching • Testing of memory hierarchy and DMA engines **Phase 3 – Application Demonstration** • Multimodal sensor data processing demonstration • Hierarchical multi-stage ML execution • Power and performance benchmarking			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for: • Design and implementation of Fibonacci ML SoC architecture • Development of ML compiler and deployment flow • System integration, validation, and benchmarking
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for: • Definition of target edge AI applications and workloads • Integration of the SoC into product demonstrators • Validation under real operating conditions

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will result in: • a validated **hierarchically scalable edge AI SoC**, • efficient processing of spatial and temporal neural networks, • and optimized power consumption across diverse workloads. For Korean SMEs, expected impact includes: • reduced development time for edge AI products, • improved energy efficiency, • and competitive differentiation in low-power AI markets.

6. References

(1) Fibonacci ML SoC

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization